

Information Technology and Knowledge Sharing in Distributed Teams

There has been a frantic pace by commercial software companies to develop and market collaboration tools to help distributed teams work more effectively. At the same time, much has been written about distributed teams. Despite all this attention, we still know very little about what are the most effective properties and features that these collaboration tools must have and under which conditions they should be utilized by distributed teams. However, in order to understand how collaboration technologies can help distributed teams perform, we need to better understand a number of things, which is what has motivated our research in this area. In particular, we have been interested in understanding how shared knowledge can help distributed teams coordinate and how this can help us influencing the design of collaboration tools:

1. Studies have shown empirical evidence that shared knowledge helps team performance (Nelson Jay G. Coopridner), but we need to learn more about the processes through which shared knowledge affects performance. It has been argued that shared knowledge of the task an of the team helps members coordinate because they can anticipate what other members are likely to do (Cannon-Bowers, Salas et al. 1993; Klimoski and Mohammed 1994). Our research has found some evidence that shared knowledge does affect coordination in asynchronous management teams (Espinosa, Carley et al. 2002) and in distributed software teams (Espinosa, Kraut et al. 2002).
2. Shared knowledge does not operate in a vacuum, but within a task context in which teams use an assortment of mechanisms, processes and tools to coordinate their work. March and Simon argued that teams communicate through such task programming mechanisms and by communicating, and that different tasks require more of one or the other depending on the uncertainties inherent in the task. This has two implications. First, the effect of shared

knowledge on coordination needs to be controlled for the presence of some of these mechanisms. This is even more important in distributed contexts in which most of the team interaction is mediated by information technologies. For example, we have found that the use of a software configuration management system is the most significant predictor of coordination success in distributed software teams (Espinosa, Kraut et al. 2002). This is so because team members use such systems not only for the intended purpose of supporting multiple developers making parallel changes to the same software product, but also to discuss issues and document software problems (Grinter 2000). And, second, shared knowledge may influence the effectiveness of some of these mechanisms. For example, it has been argued that mutual knowledge provides team members with more common ground, which makes their communication more efficient (Krauss and Fussell 1990; Cramton 2001).

3. We need to better understand how the development of shared knowledge can be fostered in a distributed team. This will enable us to design more effective tools to help teams share knowledge. Studies have found evidence that mutual trust and mutual influence have help develop shared knowledge (Nelson Jay G. Coopridner). However such mutual trust and influence cannot occur without prior team interaction (Cannon-Bowers, Salas et al. 1993; Klimoski and Mohammed 1994) and such interaction is less frequent and not very spontaneous in distributed contexts. But empirical evidence has shown that team interaction does help members develop shared task knowledge (Levesque, Wilson et al. 2001).
4. We need to better understand the differential effect of shared knowledge, and the tools that support knowledge sharing, under various distance and time separated conditions. One of the biggest problems of doing research with distributed teams is that the multiple boundaries that distribute teams often span (e.g., distance, time-zones, culture, organizational, functional) can

introduce confounds (Orlikowski 2002; Watson-Manheim, Chudoba et al. 2002; Espinosa, Cummings et al. 2003). Furthermore, supporting geographically distributed collaboration presents more substantial challenges when team members are separated by time zones (Espinosa and Carmel 2004).

In sum, effective collaboration tools for distributed teams must be ones that: (1) foster the development of shared knowledge within a team by enabling frequent, spontaneous and persistent interaction. For example, studies are showing that text-based instant messaging tools can be effective in supporting distributed team interaction [I HAVE A CITATION FOR THIS BUT IT IS IN MY OFFICE ON PAPER – I WILL GET IT LATER]. These tools also have the added benefit of providing presence awareness (Handel and Herbsleb 2002), which can be beneficial to promote mutual trust and influence; (2) enhance the ability of such shared knowledge to positively affect coordination; (3) are suited for the particular task (e.g., software configuration management systems); (4) enhance the effectiveness of other coordination mechanism and processes.

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